

Auteur: Seda Jusjajeva
 Studierichting: Informatica
 Oefeningenreeks 7B nr. 2.2

$$w = pq \sin r \text{ met } p = 2s + t, q = s - t \text{ en } r = st$$

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial p} \frac{\partial p}{\partial t} + \frac{\partial w}{\partial q} \frac{\partial q}{\partial t} + \frac{\partial w}{\partial r} \frac{\partial r}{\partial t}$$

$$\frac{\partial w}{\partial p} = q \sin r$$

$$\frac{\partial w}{\partial q} = p \sin r$$

$$\frac{\partial w}{\partial r} = pq \cos r$$

$$\frac{\partial p}{\partial t} = 1$$

$$\frac{\partial q}{\partial t} = -1$$

$$\frac{\partial r}{\partial t} = s$$

$$\frac{\partial w}{\partial t} = q \sin r - p \sin r + pqs \cos r$$

$$= (q - p) \sin r + pqs \cos r$$

$$= ((s - t) - (2s + t)) \sin st + (2s + t)(s - t)s \cos st$$

$$= (-s - 2t) \sin st + (2s^2 - st - t^2)s \cos st$$

$$= (-s - 2t) \sin st + (2s^3 - s^2t - st^2) \cos st$$

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial p} \frac{\partial p}{\partial s} + \frac{\partial w}{\partial q} \frac{\partial q}{\partial s} + \frac{\partial w}{\partial r} \frac{\partial r}{\partial s}$$

$$\frac{\partial p}{\partial s} = 2$$

$$\frac{\partial q}{\partial s} = 1$$

$$\frac{\partial r}{\partial s} = t$$

$$\frac{\partial w}{\partial s} = 2q \sin r + p \sin r + pqt \cos r$$

$$= 2(s - t) \sin st + (2s + t) \sin st + (2s + t)(s - t)t \cos st$$

$$= (2s - 2t + 2s + t) \sin st + (2s + t)(s - t)t \cos st$$

$$= (4s - t) \sin st + (2s + t)(s - t)t \cos st$$

$$\begin{aligned}
\Rightarrow Dw &= \begin{pmatrix} \frac{\partial w}{\partial t} & \frac{\partial w}{\partial s} \end{pmatrix} \\
&= ((-s-2t)\sin st + (2s^3 - s^2t - st^2)\cos st \quad (4s-t)\sin st + (2s+t)(s-t)t\cos st)
\end{aligned}$$

References